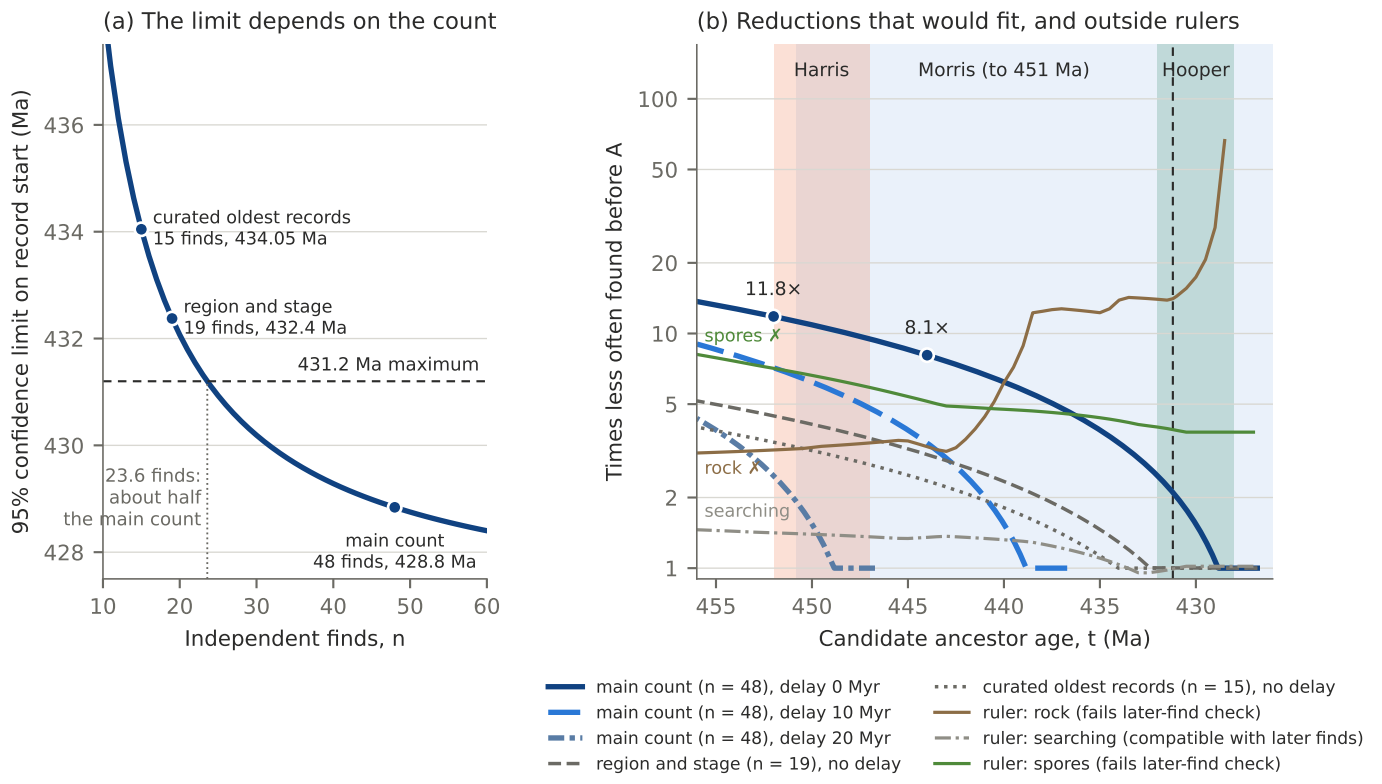
(a) The limit depends on the count
95% confidence limit on record start (Ma)
436
434
432
430
428
curated oldest records
15 finds, 434.05 Ma
region and stage
19 finds, 432.4 Ma
431.2 Ma maximum
23.6 finds:
about half
the main count
main count
48 finds, 428.8 Ma
10
20
30
40
50
60
Independent finds, n
(b) Reductions that would fit, and outside rulers
Harris
Morris (to 451 Ma)
Hooper
Times less often found before A
100
50
20
10
5
2
1
11.8×
8.1×
spores ✗
rock ✗
searching
455
450
445
440
435
430
Candidate ancestor age, t (Ma)
main count (n = 48), delay 0 Myr
main count (n = 48), delay 10 Myr
main count (n = 48), delay 20 Myr
region and stage (n = 19), no delay
curated oldest records (n = 15), no delay
ruler: rock (fails later-find check)
ruler: searching (compatible with later finds)
ruler: spores (fails later-find check)